National University of Computer and Emerging Sciences



LABORATORY MANUAL

for

Data Warehousing & Business Intelligence Lab

(DL3004)

LAB 01 • DW Foundations, PostgreSQL & Analytical SQL

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Lab Manual 01

Objectives

- Differentiate operational (OLTP) questions from analytical/DSS questions using the same retail dataset.
- Create a PostgreSQL schema, import a CSV dataset through pgAdmin, and validate that the load is complete.
- Write practical analytical SQL using filtering, aggregation, GROUP BY, HAVING, ORDER BY and Top-N patterns.
- Use date-based analysis to identify monthly sales behavior and channel/category performance.
- Use EXPLAIN ANALYZE to observe how a point lookup differs from an analytical aggregation query.
- Translate query results into concise business insights rather than reporting numbers without interpretation.

Tools	PostgreSQL, pgAdmin 4
Dataset	lab01_campuscart_sales_flat.csv
CLO Mapping	CLO 1, CLO 4
Submission	One .sql file + evidence screenshots + short insight summary

PRE-LAB: Before you start

Download the supplied CSV locally. You should be able to connect to PostgreSQL in pgAdmin and open the Query Tool. This lab assumes basic SELECT/WHERE knowledge but does not assume prior data-warehouse implementation.

1. Business Scenario

CampusCart is a university-focused online and kiosk retailer. Management has transactional sales data for 2025 but no analytical workflow. Your job is to load the data into PostgreSQL, answer decision-support questions, and explain what the results mean for management.

> **CONCEPT: Think like an analyst**
>
> A transactional system answers questions such as “What happened to order 104512?” A decision-support workload asks “Which categories, channels and regions drove revenue this semester?” You will run both kinds of queries today.

2. Dataset Overview

The dataset contains 7,800 synthetic order-line records from January to December 2025. Returned orders have negative net revenue and cancelled orders have zero net revenue.

Column	Type	Meaning
order_id	integer	Unique order-line identifier
order_date	date	Transaction date
customer_id	integer	Customer identifier
customer_segment	text	Student, Faculty, Alumni or Staff
product_id	integer	Product identifier
product_name	text	Human-readable product name
category	text	Product category
brand	text	Product brand
region_id	text	Region code
region_name	text	Sales region
channel	text	Web, Mobile App or Campus Kiosk
quantity	integer	Units in the order line
unit_price_pkr	numeric	Selling price per unit
discount_pct	integer	Discount percentage applied
payment_method	text	Payment channel
order_status	text	Completed, Returned or Cancelled
gross_revenue_pkr	numeric	Revenue before discount/status adjustment
net_revenue_pkr	numeric	Net amount; returns are negative, cancellations zero

3. Environment Setup & CSV Import

Create a dedicated schema and a table that matches the CSV. The DDL below is starter infrastructure, not the answer to the analytical questions.

Starter DDL

```
CREATE SCHEMA IF NOT EXISTS lab01;

CREATE TABLE lab01.campuscart_sales (
    order_id            INT,
    order_date          DATE,
    customer_id         INT,
    customer_segment    TEXT,
    product_id          INT,
    product_name        TEXT,
    category            TEXT,
    brand               TEXT,
    region_id           TEXT,
    region_name         TEXT,
    channel             TEXT,
    quantity            INT,
    unit_price_pkr      NUMERIC(12,2),
    discount_pct        INT,
    payment_method      TEXT,
    order_status        TEXT,
    gross_revenue_pkr   NUMERIC(14,2),
    net_revenue_pkr     NUMERIC(14,2)
);
```

- In pgAdmin, refresh Schemas > lab01 > Tables after running the DDL.
- Right-click lab01.campuscart_sales > Import/Export Data > Import.
- Choose the CSV, set Header = Yes, Delimiter = comma, Encoding = UTF-8, then import.
- Do not continue until the row count and basic date range look plausible.

CHECKPOINT: Load validation

Record: total rows, minimum order_date, maximum order_date, and the number of distinct order_status values. If these do not match the data description, fix the import before continuing.

4. Data Familiarization & Profiling

#	Task	Evidence
1	Return the total row count and number of distinct customers.	Query result
2	List every distinct category, channel and order_status.	Query result
3	Find minimum, maximum and average net_revenue_pkr. Explain why the minimum can be negative.	Result + 1 sentence
4	Count records by order_status and sort from highest to lowest.	Query result
5	Check whether any quantity is zero/negative or any order_date falls outside 2025.	Validation result

HINT: SQL reminder

For analytical queries, think in the order: filter rows -> group rows -> calculate aggregate -> filter groups -> sort/present. Relevant clauses are WHERE, GROUP BY, HAVING and ORDER BY.

5. Guided Analytical SQL

5.1. Revenue & Order Performance

#	Task	Evidence
1	Calculate total net revenue, total gross revenue and total quantity for the full year.	1 result row
2	Calculate monthly net revenue. Sort chronologically, not alphabetically.	12 rows
3	Find the Top 5 products by net revenue. Include product_name and category.	5 rows
4	Compare net revenue by channel and calculate the number of order lines in each channel.	3 rows
5	Return categories whose net revenue exceeds PKR 1,000,000 using HAVING.	Filtered grouped result

5.2. Decision-Support Questions

Write one query for each management question. Your output should be easy for a non-technical manager to read. Use aliases and sensible sorting.

#	Task	Evidence
1	Which three region/category combinations generated the highest net revenue?	Top 3 result
2	Which customer segment generated the highest average order-line value among Completed orders only?	Grouped result
3	Which month had the highest return count, and what percentage of that month's records were returns? You may use two queries if needed.	Result + calculation
4	Compare average discount_pct across channels. Does the highest-discount channel also have the highest net revenue?	Result + answer
5	Management wants a "high-value" list. Return Completed order lines above PKR 10,000 net revenue, newest first.	Detail result

COMMON PITFALL: Do not confuse gross and net

For performance questions, use net_revenue_pkr unless the question explicitly asks for gross revenue. Returns intentionally reduce net revenue.

6. OLTP vs DSS: Observe the Workload Difference

6.1. Run Two Different Query Shapes

Use EXPLAIN ANALYZE before each query. Do not create indexes in this lab; indexing is taught later.

Experiment queries

```
-- Query A: operational / point-style lookup
EXPLAIN ANALYZE
SELECT *
FROM lab01.campuscart_sales
WHERE order_id = 104250;

-- Query B: analytical / DSS-style aggregation
EXPLAIN ANALYZE
SELECT date_trunc('month', order_date) AS month,
       category,
       SUM(net_revenue_pkr) AS net_revenue
FROM lab01.campuscart_sales
WHERE order_status <> 'Cancelled'
GROUP BY date_trunc('month', order_date), category
ORDER BY month, net_revenue DESC;
```

#	Task	Evidence
1	Record the top plan node and actual execution time for Query A.	Plan evidence
2	Record the top plan node and actual execution time for Query B.	Plan evidence
3	In 2-3 sentences, explain why Query B is representative of DSS/BI work while Query A is closer to OLTP access.	Short explanation

7. Applied Challenge - Management Brief

7.1. Build a 4-Query Evidence Pack

You are given four decisions. For each, write a query and state one evidence-based recommendation. There is no single “correct” wording, but your SQL and interpretation must agree.

#	Task	Evidence
1	Inventory: identify two products that deserve extra stock before the next semester peak. Justify using revenue and/or quantity.	SQL + 1 insight
2	Marketing: identify one channel that should receive additional promotional budget.	SQL + 1 insight
3	Operations: identify one region with a comparatively high return count.	SQL + 1 insight
4	Pricing: identify a category where heavier discounting does not appear to translate into stronger net revenue.	SQL + 1 insight

ASSESSMENT: Evidence over opinion

A recommendation without a query result is not an analytical answer. Screenshots should show enough of the result to verify the conclusion.

8. Submission Requirements

- Lab01_<RollNo>.sql containing all validation, analytical and EXPLAIN ANALYZE queries in task order.
- A PDF or document containing screenshots for the required checkpoints and the 4-query management brief.
- At the end of the SQL file, add a block comment titled “Key Insights” with 3 concise findings from the dataset.
- Do not submit copied outputs without the SQL that produced them.

End of Lab Manual 01

Good Luck!!!